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AI-Based Recognition of Fruit and Vegetable Spoilage: Towards Household Food Waste Reduction

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Abstract

Food waste is a critical global issue, with approximately one-third of produced food being wasted annually. Indonesia, contributing around 20.93 million tonnes to this waste, faces several economic and environmental impacts. The primary causes of food waste in Indonesian households include a lack of knowledge about food storage and expiration dates, along with the unreliability of traditional freshness assessment methods. This research will evaluate three pre-trained AI models – MobileNetV2, VGG19, and EfficientNetV2S – using the 'Fresh and Rotten Classification' dataset from Kaggle, which contains 30.4k images depicting both fresh and rotten produce, in hopes to extend this research to integrate these models to AI-based tools to recognize spoilage in produce, thereby reducing household food waste. EfficientNetV2S emerged as the most effective model, achieving a 97.61% accuracy and a 97.59% F1-score, indicating robust performance and suitability for real-time applications in households. Despite promising results, challenges such as dataset quality and class imbalance were noted. Future research should focus on improving models' performance through hyperparameter tuning, transfer learning, and developing comprehensive datasets. Integrating these models into user friendly applications could significantly contribute to reducing food taste and promoting sustainable consumption habits in Indonesia.

Keywords: Food waste; Fruits; Classification; VGG19; EfficientNetV2S; MobileNetV2

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1. Introduction

Food waste is a paramount global challenge that threatens the environment, and economy and has significant social effects [1]. According to the United Nations Environment Programme's (UNEP's) Food Waste Index Report 2021, people globally waste 1 billion tonnes of food annually [2]. This amounts to about one-third of food produced globally being wasted or lost. The food waste problem is critical, especially in Indonesia, where according to the United Nations Environment Programme's (UNEP's) Food Waste Index Report 2021, Indonesia produces about 20.93 million tonnes of food waste annually, equivalent to the loss of IDR 231 - IDR 551 trillion per year, putting Indonesia in second place for being the country with the most production of food waste [2].

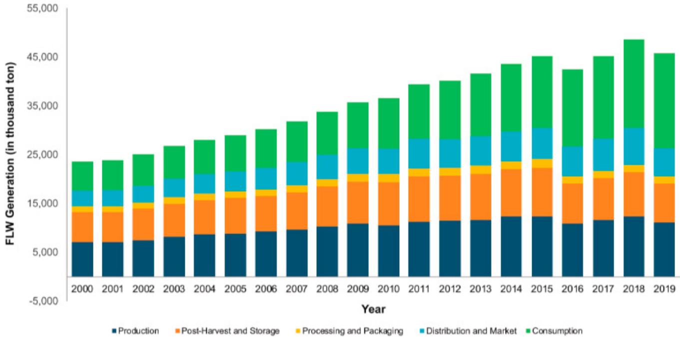

Fig. 1. Food Loss and Waste Generation in Indonesia's Food Supply Chain from 2000 to 2019 [3]

Despite the increasing concern about food waste in Indonesia, food workers, and consumers often struggle to identify and manage food spoilage. Lack of knowledge about food storage, handling, and expiration dates leads to premature disposal of edible items, particularly fruits and vegetables, which contributes to a significant portion of household food waste in Indonesia [3].

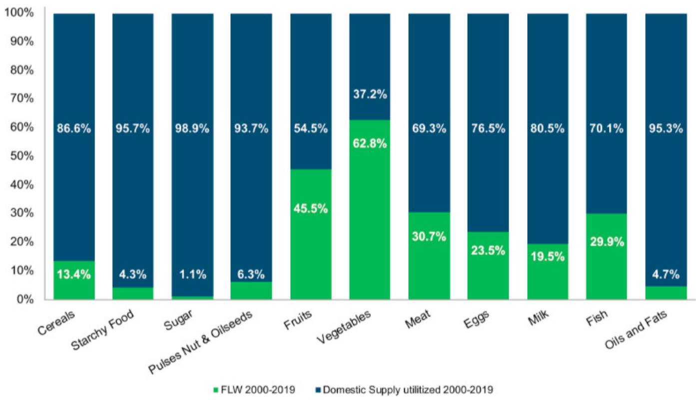

Fig. 2. Food Loss and Waste Generation in Indonesia's Food Supply Chain from 2000 to 2019 [3]

Traditional methods of assessing food freshness, such as visual inspection by performing manual checks, can be subjective and unreliable. In addition, there is also a lack of real-time monitoring tools that could assist consumers in tracking shelf life and detecting food spoilage accurately [3]. Therefore, heightens the urgency for an efficient and user-friendly solution to minimize food waste at household levels in Indonesia.

This research's main objective is to develop an AI-based model for the recognition of fruit and vegetable spoilage, with the ultimate goal of creating a user-friendly recognition tool for consumers at the household level. The envisioned tool will enable consumers to make informed decisions on food consumption, assisting in reducing household food waste, and leading to environmental conservation and cost savings. Additionally, the tool has the potential to promote more sustainable consumption habits and contribute to the broader goal of achieving a more resilient and equitable food system.

This research will focus on the training and evaluation of three different pre-trained AI models: MobilenetV2, VGG19, and EfficientnetV2S. These models will be trained on the 'Fresh and Rotten Classification' dataset from Kaggle [4], comprising 30.4k images of fruits and vegetables, depicting both fresh and rotten states. The ultimate goal is to develop a robust AI-based tool capable of accurately recognizing fruit and vegetable spoilage at the household level in Indonesia. Through thorough training and evaluation of these models, this research aims to create a scalable and adaptable solution that can effectively assist consumers in reducing household food waste and promoting sustainable consumption habits.

2. Literature Review

Previous studies have also leveraged AI to differentiate fresh and spoiled fruits and vegetables. For instance, Convolutional Neural Networks (CNNs) were often used for image processing and extracting key features to classify fruits and vegetables [5]. Another study conducted utilised YOLOv2 algorithm that outperformed most models used to identify and classify fruits and vegetables. YOLOv2 was able to detect image features faster and it was easier to train its models with the provided data [6]. One study based on CNNs compared eight pre-trained deep learning which include AlexNet [7], GoogLeNet [8], ResNet18, ResNet50, ResNet101, VGG16, VGG19, and NasNetMobile. The study concluded that after conducting evaluation on each of the models with the same dataset, it was discovered that VGG19 was the most accurate model displaying 91.50% accuracy [9].

Across the reviewed literature, Convolutional Neural Networks (CNNs) emerge as a pivotal technology for fruit quality assessment and freshness detection. Studies consistently highlight the robustness and effectiveness of CNNs in learning complex patterns and generalizing well to unseen samples, offering advantages in terms of speed and accuracy compared to other models [6].

Additionally, various feature extraction techniques have been explored, including Histogram (Hist), Gray Level Co-occurrence Matrix (GLCM), Bag of Features (BoF) [7], and CNN-based features, with the latter consistently proving successful in analyzing complex food matrices [10]. The application of transfer learning, particularly leveraging pre-trained CNN models, is shown to be effective for fruit and vegetable quality assessment, enabling successful classification tasks without requiring extensive datasets for training [9]. Model comparison and evaluation across studies involve a range of architectures such as ResNet [11], VGG, MobileNetV2, and YOLOV2, emphasizing metrics like accuracy, recall, and F1-score, alongside considerations for computational efficiency and energy consumption [6,9,12].

Comparative analyses demonstrate the superior performance of CNNs compared to other algorithms, such as KNN, ANN, Fuzzy, SVM, and DT, in tasks like lemon classification based on apparent imperfections, as highlighted in the study on sour lemons. The study also underscores the potential of improved CNN architectures in combination with image processing methods to enhance traditional grading methods for sour lemons, preserving product quality, improving marketability, and contributing to waste management [13]. Similarly, in detecting defective apples and external defects on tomatoes, CNN-based architectures have shown remarkable success. The study on defective apples illustrates the integration of a CNN-based classification model into a fruit sorting system, achieving high accuracy and demonstrating potential for commercial implementation [14].

The paper on persimmon fruits indicates that CNN models can successfully classify rapid over-softening fruits with high accuracy using only normal picture images of the outer appearance. Moreover, the paper highlights the correlation between prediction values in the classification and the days to fruit softening, suggesting the potential for predicting fruit shelf life in persimmons [15]. Meanwhile, the paper on tomatoes introduces a deep neural network binary classifier for detecting external defects, achieving state-of-the-art performance, and demonstrating the extensibility of the model to other foods [16]. The collective findings underscore the significance of continuous improvement and innovation in this field, emphasizing the importance of refining CNN models, expanding datasets, and exploring real-time implementation for practical applications.

Several challenges and research gaps become apparent after reviewing the papers. Firstly, there is a notable issue regarding the diversity and size of datasets used for training AI models. While some studies mention data augmentation techniques to increase dataset size [13], the availability of diverse and large-scale datasets remains a challenge, particularly for less common fruits or vegetables. Secondly, while Convolutional Neural Networks (CNNs) are widely used and effective, there might be challenges in model generalization across diverse types of produce. Many papers focus on specific fruits like lemons, apples, or tomatoes, indicating a need for research demonstrating the scalability and generalizability of these models to a broader range of produce. There is also a need for standardization in evaluation metrics and benchmark datasets for comparing different AI models in fruit quality assessment. Standardized evaluation frameworks would facilitate fair comparisons and enable researchers to identify the strengths and weaknesses of various approaches more effectively.

Furthermore, the practical implementation of AI-based quality assessment systems in household settings may pose challenges related to affordability, accessibility, and user-friendliness. These factors could affect the adoption of such systems by consumers and, consequently, their impact on reducing household waste.

While Convolutional Neural Networks (CNNs) show great promise in fruit quality assessment, there are significant challenges to address, including dataset diversity, model generalization, and standardization of evaluation metrics. Additionally, the practical implementation of AI-based systems in household settings may face obstacles related to affordability and user-friendliness. Overcoming these challenges will require ongoing innovation, collaboration, and investment to realize the full potential of AI-driven solutions in reducing household waste and enhancing fruit quality assessment.

3. Methodology

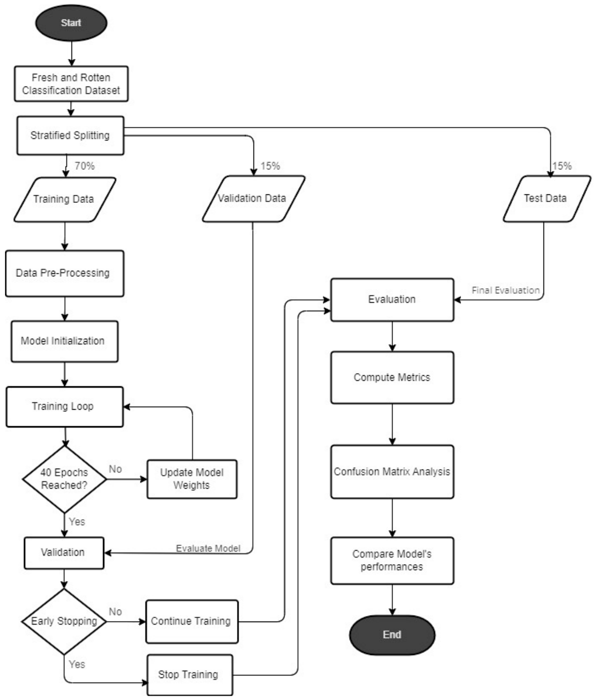

Fig. 3. Flowchart representing the methodology

3.1. Datasets

In the fruits and vegetables classification model, the dataset used is taken from Kaggle named "Fresh and Rotten Classification" which is composed of 30,357 images of fresh and rotten fruits and vegetables such as apples and bananas [4]. The dataset has been labelled as 'fresh' and 'rotten' for each class of fruits and vegetables. The nine classes include apples, bananas, bitter gourd, capsicum, cucumber, okra, oranges, potatoes, and tomatoes. Additionally, the dataset has been pre-augmented with random rotations, horizontal and vertical flipping, and cropping. Furthermore, the dataset has initially been split into a 78% training set and 22% testing set. This research re-splits the dataset and performs stratified splitting to ensure that the class distribution will be able to represent the class distribution of the entire dataset, the dataset is split into a 70% training set, 15% validation set, and 15% testing set of the proportion of images from each class respectively.

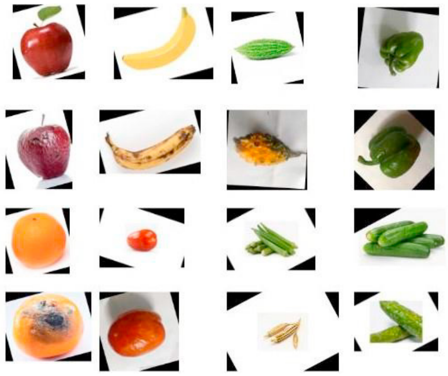

Fig. 4. representing the sample image from the “Fresh and Rotten Classification”

3.2. Methods

3.2.1. MobileNetV2

MobileNetV2 was selected for this research due to its efficiency and effectiveness in multiclass image classification, particularly in resource-limited environments. MobileNetV2 employs an architecture optimized for mobile and embedded vision applications, defined by the usage of depth wise separable convolutions and inverted residuals [17]. These designs significantly reduce the number of parameters and computational load while preserving high accuracy. This research will utilize the standard variant, MobileNetV2 1.0., which balances performance and efficiency, making it ideal for real time classifications tasks on devices with limited computational resources. By utilizing MobileNetV2, this research aims to provide a viable solution for households that can run efficiently on common consumer devices such as smartphones and tablets.

3.2.2. VGG19

VGG19 is selected for its well-established performance and simplicity in image classification tasks. VGG19 excels in learning highly discriminative features, due to its architecture that consists of deep stacks of convolutional layers with small receptive fields, which allow the network to capture fine features. VGG19’s depth comprises of 19 weight layers (16 convolutional and 3 fully connected layers which enables it to model complex patterns in data) [18]. This research will utilize the standard pre-trained version of VGG19. The standard pre-trained version of VGG19 will be utilized in this research for its robustness which can ensure the identifications of subtle differences in produce quality. This feature makes VGG19 a strong candidate for systems requiring detailed and precise image analysis, despite its higher computational demands making it suitable for fruits and vegetables freshness identification.

3.2.3. EfficientNetV2S

EfficientNetV2S is selected for this research due to its performance in both accuracy and efficiency. EfficientNetV2S, the small variant of EfficientNetV2, will be utilized for its optimized balance between computational efficiency and high accuracy. This model’s superior performance is the result of employing a combination of neural architecture search (NAS) and compound scaling to simultaneously scale network depth, width, and resolution [19]. This approach ensures that the model can achieve high accuracy without the need for extensive computational resources, making it suitable for real-time applications in household environments. EfficientNetV2S’ ability to deliver high performance with fewer computational demands makes it applicable for developing user-friendly applications designed to reduce food waste through real-time food detection.

3.3. Training Procedure

3.3.1. Data preprocessing

This step involves pre-processing the image data to prepare it for training and ensures consistency across the images. All images will be resized to 224 x 224 pixels for the MobileNetV2, EfficientNetV2S and VGG19. Pixel intensity normalization will be applied and will scale the values to a range between 0 and 1, which assists the model in acquiring faster convergence during the training phase. This will be done with Z-score normalization.

$$Z = \frac{x - \mu}{\sigma} \tag{1}$$

where x is the raw score of the pixel intensity, μ is the mean intensity value across all pixels in the image and σ is the standard deviation of the intensity across all pixels in the image.

Augmentation techniques such rescaling, augmenting brightness range, rotating, width shifting, height shifting, shearing, flipping, and filling will be implemented to enhance the dataset and improve the performance robustness of the models.

3.3.2. Model initialization

This research employs transfer learning, a technique that uses pre-trained models to accelerate the training process and leverage the learned features for classifying fresh and rotten fruits and vegetables. The pre-trained versions of MobileNetV2 1.0, VGG19, and EfficientNetV2S are chosen for their efficiency, accuracy, and compatibility with various computational environments and will be utilized within Visual Studio Code with NVIDIAGeForce RTX 4070 Laptop GPU with 16 GB RAM and Python 3.11.17 kernel.

The initial layers (base network) of these pre-trained models will be frozen to preserve the knowledge collected within the layers, allowing reusability without modifying the current classification task [20]. The final classification layer is not suitable for multiclass classification and will be replaced with a new dense layer with multiple output neurons, where each fruit and vegetable class (e.g., apples, oranges, okra, etc) has two neurons: one for fresh and one for rotten. These neurons will then be coupled with a SoftMax activation to produce a probability interpretation ofthe output [21].

$$\mathrm{f}(\mathrm{x_i}) = \frac{\mathrm{e}^{\mathrm{x_i}}}{\sum_{\mathrm{j}=1}^{\mathrm{N}} \mathrm{e}^{\mathrm{x_j}}} \; (\mathrm{i} = 1, 2, \ldots, \mathrm{N}) \tag{2}$$

where x is the input factor containing raw scores from the last layer of the neural network, i is the i-th element of the input factor x, N is the total number of classes.

The pre-trained weights from the chosen versions of MobileNetV2 1.0, VGG19, and EfficientNetV2S will then be loaded into the corresponding layers to enable the model to prioritize adapting the final layers to classify fresh and rotten fruits and vegetables. All three pre-trained models will utilize the Adam optimizer, with varying learning rates.

3.3.3. Training Loop

This process involves iterating the training of the three pre-trained CNN models to classify fresh and rotten vegetables and produce. The loop begins by initializing the models with pre-trained weights. With each loop, the models process a batch of images, predict their labels, and compute the loss based on the discrepancy between the predicted and actual labels using categorical cross-entropy as the loss function. The optimizer of each model then updates the model weights to decrease the loss, utilizing gradients calculated through backpropagation. This iterative process continues for 30 epochs, during which the models are fine-tuned to the classification task.

3.3.4. Validation

After 30 epochs, the models will be evaluated using the validation set, and the validation loss will be computed. Throughout the training process, techniques such as early stopping will be used to supervise the validation accuracy and prevent overfitting [22]. Additionally, learning rate scheduling will be utilized to enhance convergence speed. When validation loss stops improving for a specified number of epochs, training will be stopped [23]. This validation step ensures that the models do not overfit the training data and can generalize to new, unseen data.

3.3.5. Evaluation

After the training process, the models will be evaluated on the testing set. Metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis will be used to evaluate the model's performance on the testing set. Accuracy analysis measures the proportion of correctly classified images across all classes (fresh and rotten for each type of fruit and vegetable). The confusion matrix will visualize the model's performance across all classes, showing how many images were correctly and incorrectly classified for each type of fruit and vegetable in both fresh and rotten states [24]. This evaluation will help in identifying the most suitable model for specific applications and potentially guide further improvements.

4. Results and Discussion

Table 1. Table of results of the CNN models

Models	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
MobileNetV2	95.61	96.41	95.05	95.74
VGG19	87.05	94.22	74.09	82.95
EfficientNetV2S	97.61	97.78	97.41	97.59

Among the results obtained by each model, the EfficientNetV2S model displayed the highest value in all metrics achieving 97.61% in accuracy, 97.78% in precision, 97.41% in recall and an F1-score of 97.59%. MobileNetV2 also showed promising results with an F1-score of 95.74% and an accuracy score of 95.61%.

4.1. MobileNetV2

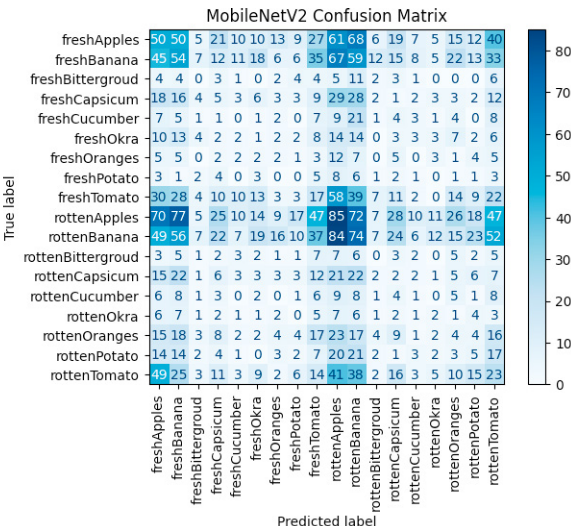

Fig. 5. Confusion matrix of MobileNetV2

Based on Fig.5. The model is best at predicting for rotten apples since it scored the highest value of 85 compared to other labels. This implies that out of 630 predictions made for rotten apples, about 13.49% were correct. However,the model predicted poorly on determining the freshness of vegetables.

4.2. VGG19

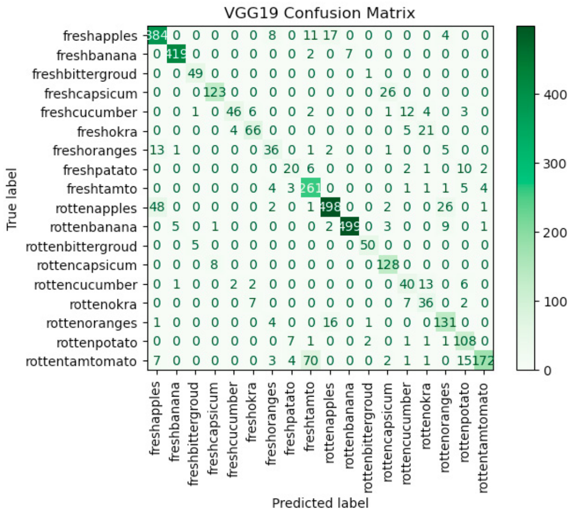

Fig. 6. Confusion matrix of VGG19

The model was able to predict and identify more accurately compared to the MobileNetV2 model, by scoring the highest when determining whether the image of apples and bananas were fresh or rotten as evident in Fig.6. For the vegetable's dataset, the model performed best when predicting and identifying fresh and rotten capsicums scoring 123 and 128 respectively.

4.3. EfficientNetV2S

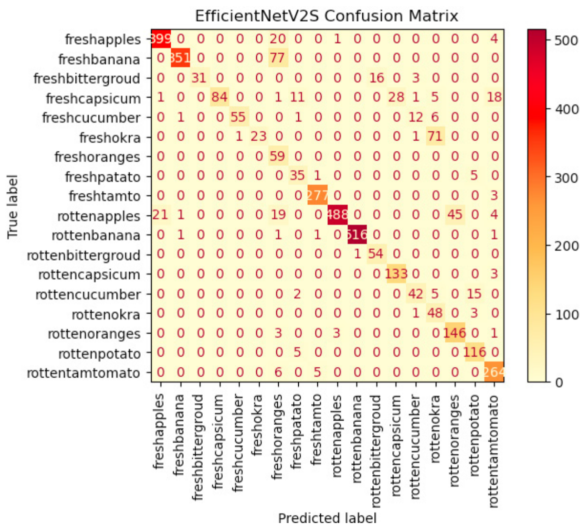

Fig. 7. Confusion matrix of EfficientNetV2S

The model was able to predict and identify more accurately for the fruit's dataset compared to the VGG19 model. The EfficientNetV2S model scored a peak value of 399 and 488 for fresh and rotten apples and 351 and 516 for fresh and rotten bananas as evident in Fig.7. However, there is a decline in performance when predicting and identifying for fresh and rotten vegetable's dataset, where for labels of fresh and rotten capsicums scoring 84 and 133, respectively.

5. Conclusion

Food waste has been proven to pose a detrimental challenge to Indonesian's environment and economy. One of the main reasons for this is due to the inconsistency of criterion to categorise whether the food, particularly fruits and vegetables, are edible. Thus, our study leveraged on CNN models namely MobileNetV2, VGG19 and EfficientNetV2S to help aid individuals determine the freshness of fruits and vegetables and ultimately other types of food. Based on our results, EfficientNetV2S achieved the highest F1-score of 97.59% and 97.61% accuracy score, proving robust performance across classes compared to the other models. Its lightweight architecture ensures faster inference times and low resource consumption makes it suitable for real-time applications in household environments. It also allows for the development of lightweight applications and can be easily integrated into smartphones or IoT devices due to their efficient architectures. This integration aligns with the common technological landscape in many Indonesian household where smartphones are common. The dataset used was limited to only some common Indonesian produce (e.g. banana, cucumber, potato), which may not fully represent the diversity of produce available in the Indonesian households. There was a significant imbalance between the number of fruit images and vegetable images, with fruit images being more prevalent. This imbalance led to biased model performance, favoring the classification of fruits over vegetables. Future work should focus on fine-tuning hyperparameters including adjusting learning rates, batch sizes and dropout rates to optimize model accuracy and reduce overfitting. Employing transfer learning can also significantly improve model performance. Furthermore, we hope that future research can extend this research by exploring various ways of integrating enhanced models into household applications and creating suitable datasets for the development of enhanced models.

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